

# The Missing Premise in Baryogenesis: The Scope of the Zero Antecedent

## 1. The observed asymmetry

The observable universe contains a small but unmistakable excess of baryonic matter over antibaryonic matter. A convenient quantity for stating the distinction is the net baryon asymmetry

$$\Delta_B = n_B - n_{\bar{B}},$$

where  $n_B$  and  $n_{\bar{B}}$  are the baryon and antibaryon number densities. At epochs when the surviving antibaryon abundance is negligible, the same asymmetry is commonly expressed through a normalized baryon-to-photon or baryon-to-entropy ratio. The observed value is small: the baryon-to-photon ratio is of order  $6 \times 10^{-10}$ . Big-Bang nucleosynthesis and cosmic-microwave-background determinations independently constrain the cosmic baryon abundance and agree closely within their respective uncertainties.<sup>1</sup>

This observational fact creates the familiar baryogenesis problem. A universe containing nearly equal numbers of baryons and antibaryons would leave little surviving ordinary matter after annihilation. The observed positive baryon excess therefore invites an account of how the asymmetry arose. Mechanisms proposed for that job include heavy-particle decays, electroweak baryogenesis, leptogenesis, Affleck–Dine dynamics, and other uses or extensions of early-universe particle physics. For mechanisms that dynamically generate a net baryon asymmetry from an initially baryon-symmetric state, the usual Sakharov conditions identify the necessary ingredients: baryon-number violation, C and CP violation, and departure from

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<sup>1</sup> Planck Collaboration, N. Aghanim et al., “Planck 2018 Results. VI. Cosmological Parameters,” *Astronomy & Astrophysics* **641**, A6 (2020), doi:10.1051/0004-6361/201833910; R. J. Cooke, M. Pettini, and C. C. Steidel, “One Percent Determination of the Primordial Deuterium Abundance,” *Astrophysical Journal* **855**, 102 (2018), doi:10.3847/1538-4357/aaab53. Planck finds  $\Omega_{\text{bh}}^2 \simeq 0.0224$  from the CMB, while the deuterium-based BBN determination provides an independent baryon-density inference consistent with the CMB result.

thermal equilibrium, subject to the qualifications appropriate to the mechanism under study.<sup>2</sup>

That program is physically well motivated, but it contains two logically distinct questions. The first is observational: why does the late universe have  $\Delta_B \neq 0$ ? The second concerns the antecedent from which the relevant physical history began: was  $\Delta_B = 0$  before the baryon-asymmetry-changing dynamics under consideration?

The first question does not answer the second.

In particular,

$$\Delta_B(\text{late}) \neq 0$$

does not imply

$$\Delta_B(\text{antecedent}) = 0.$$

The late asymmetry establishes the state that a successful cosmological history must reach. It does not, by itself, specify the baryon asymmetry on an earlier boundary of that history. A zero antecedent may be a physically motivated initial condition, the result of inflationary dilution, the output of a thermal or washout regime, or an explicit assumption of a particular baryogenesis model. Each of those possibilities can be examined on its own terms. None follows simply from observing nonzero baryon asymmetry later.

This distinction matters because “the origin of the baryon asymmetry” and “the generation of baryon asymmetry from zero” are not identical problems. They coincide only when the relevant zero antecedent has itself been established at the scope of the claim.

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<sup>2</sup> A. D. Sakharov, “Violation of CP Invariance, C Asymmetry, and Baryon Asymmetry of the Universe,” *JETP Letters* **5**, 24–27 (1967); V. A. Kuzmin, V. A. Rubakov, and M. E. Shaposhnikov, “On Anomalous Electroweak Baryon-Number Non-Conservation in the Early Universe,” *Physics Letters B* **155**, 36–42 (1985), doi:10.1016/0370-2693(85)91028-7; M. Fukugita and T. Yanagida, “Baryogenesis without Grand Unification,” *Physics Letters B* **174**, 45–47 (1986), doi:10.1016/0370-2693(86)91126-3; I. Affleck and M. Dine, “A New Mechanism for Baryogenesis,” *Nuclear Physics B* **249**, 361–380 (1985), doi:10.1016/0550-3213(85)90021-5. These establish the classic dynamical conditions and representative electroweak, leptogenesis, and scalar-condensate routes invoked in the opening discussion.

## 2. The hidden antecedent

A generation mechanism answers a transition question. It describes how one physical state can evolve into another under specified interactions, fields, boundary conditions, and cosmological dynamics. When a baryogenesis calculation begins with vanishing net baryon asymmetry and ends with  $\Delta_B \neq 0$ , it can therefore answer a well-defined question: how can this model produce the observed kind of asymmetry from this model's symmetric antecedent?

The additional step from that result to a statement about the antecedent of the observable universe is a scope step.

A zero imposed on a model's initial surface is a statement about that model. A zero reached after washout is a statement about histories in which the relevant washout operates with the required strength. A zero inferred after inflationary dilution is a statement about the asymmetry, carrier, and inflationary history to which the dilution argument applies. Thermal-equilibrium relations constrain the charge combinations exposed to the relevant equilibrium reactions. None of these statements becomes broader merely because the associated mechanism is standard, successful, or widely studied.

The relevant physical object must therefore be kept explicit. Net baryon number  $B$  is not interchangeable with  $B - L$ , lepton number, flavor asymmetries, chemical potentials, or charges stored in other fields or sectors. Those quantities can be physically coupled: an asymmetry initially stored in one sector may later be transferred into baryon number, and electroweak sphaleron processes can reprocess baryon and lepton charges. But a statement that one of those neighboring quantities vanishes—or does not vanish—is not automatically a statement about  $\Delta_B$ . The distinction becomes especially important when asking what, precisely, an early-universe process has erased and what information may survive it.

Time introduces a second scope problem. A cosmological model can begin its calculation on a surface that already contains a nonzero baryon asymmetry. That alone does not establish that the asymmetry was antecedent in the broader sense relevant here. An unmodeled earlier interval could contain the very zero-to-nonzero transition at issue. Conversely, beginning a calculation on a symmetric surface does not establish that every physically admissible history shares that surface. The antecedent boundary

has to be part of the physical claim, not chosen retrospectively to produce the desired answer.

For this reason, the relevant comparison is not between “models that assume zero” and “models that assume nonzero.” It is between physical histories whose scopes are broad enough to include the antecedent boundary being tested and whose later evolution remains compatible with the established cosmological record.

This immediately changes the evidentiary burden. To claim that generation from zero is required across a class of histories, it is not enough to show that zero is convenient, natural, strongly diluted toward, or adopted by successful mechanisms. The zero antecedent has to hold across that class at the quantity, epoch, and physical scope for which the generation claim is being made.

The strongest version of the question can therefore be stated without choosing a baryogenesis mechanism:

**Do all physically admissible histories compatible with the observed cosmological record share a zero net-baryon antecedent at the relevant boundary?**

A positive answer would make generation from zero a live burden across that entire comparison class. A negative answer would require only one physically admissible history in which the relevant antecedent net baryon asymmetry is nonzero. Failure to establish either result would leave the antecedent question unresolved at that scope.

That is the narrower problem examined here. It does not ask whether known baryogenesis mechanisms can produce the observed asymmetry; many can under their stated assumptions. It asks whether the zero antecedent from which such production is often calculated can itself be extended from restricted physical histories to the full class of histories still compatible with what the universe presently allows us to describe.

### **3. The scope of the zero premise**

The scope at issue is not simply spatial. A statement about the “universe” can refer to the observable cosmological domain, to an assumed global extension beyond the

horizon, or to a class of possible histories compatible with the observations available within that domain. Those are different claims. The question considered here is principally a history-space question: whether every physically admissible history capable of describing the observed cosmological record shares the relevant zero-baryon antecedent. It should not be read as an empirical statement about every causally inaccessible region of a larger cosmos.

Let  $H_{\text{adm}}$  denote that admissible comparison class. Membership cannot depend on whether a history begins with  $\Delta_B = 0$ , because doing so would build the desired answer into the comparison. A relevant history must instead be physically articulated, internally coherent under the dynamics it invokes, broad enough to include the antecedent boundary under examination, and compatible with the observational envelope at the epochs it claims to describe. In practice, that envelope includes the baryon abundance inferred independently from BBN and the CMB, applicable constraints on primordial perturbations and entropy evolution, and any particle-physics or relic constraints triggered by the particular history.<sup>3</sup>

$H_{\text{adm}}$  is an intensional, evidence-responsive comparison class rather than a finite catalogue of models. Its membership can change as theory and observation change. The result developed here concerns the present physical coverage of that class; it does not assert that universal-zero coverage is undecidable in principle.

This criterion deliberately does not require every admissible history to contain inflation, a particular reheating process, or a conventional baryogenesis mechanism. Those structures can impose strong constraints when a history invokes them, but making them universal admission conditions would confuse a successful model family with the full question being tested. For the same reason, the Sakharov conditions remain central to mechanisms that claim dynamical production of a baryon asymmetry in the regime to which they apply, but they cannot be used to exclude a history merely because that history treats an asymmetry as inherited boundary data. Doing so would presuppose that generation from zero is mandatory while ostensibly testing whether it is.

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<sup>3</sup> R. J. Cooke, M. Pettini, and C. C. Steidel, *Astrophysical Journal* **855**, 102 (2018), doi:10.3847/1538-4357/aaab53; Planck Collaboration, N. Aghanim et al., *Astronomy & Astrophysics* **641**, A6 (2020), doi:10.1051/0004-6361/201833910; Planck Collaboration, Y. Akrami et al., “Planck 2018 Results. X. Constraints on Inflation,” *Astronomy & Astrophysics* **641**, A10 (2020), doi:10.1051/0004-6361/201833887. These supply the principal BBN/CMB baryon-density and primordial-perturbation constraints used here to define the observational envelope rather than the antecedent itself.

A second distinction concerns what “zero” means. Exact zero is the statement

$$\Delta_B = 0.$$

Dilution to a negligibly small value is not mathematically the same statement. Nor is late-time insensitivity to an initial condition. Either may be sufficient for a particular physical calculation, but an “effective zero” requires an operational threshold: small compared with what, at what epoch, and for what downstream observable? Without that threshold, strong suppression can silently become exact erasure in the language of the argument.

This matters because the strongest claim has a demanding form. If generation from zero is to be required across  $H_{\text{adm}}$ , then the relevant antecedent must satisfy

$$\forall h \in H_{\text{adm}}, \Delta_B[C_B(h)] = 0,$$

where  $C_B(h)$  denotes the antecedent boundary appropriate to history  $h$ . For each history,  $C_B(h)$  is fixed from that history’s stated dynamics and boundary conditions before the value or sign of  $\Delta_B$  at the boundary is used in adjudication. The notation does not require every candidate cosmology to share one universal clock time. It requires each history to cover the causal interval relevant to its own account of the observed baryon asymmetry, so that its starting boundary cannot simply be placed after an unmodeled zero-to-nonzero transition.

This also prevents a superficially easy counterexample. A model that begins at some early time with  $\Delta_B \neq 0$  has not, merely by choosing that starting surface, shown that the antecedent relevant to the broader question was nonzero. The asymmetry could have been generated earlier. A genuine counterexample must either carry the nonzero state as boundary data at the appropriate comparison scope or physically describe the earlier history by which it reaches that boundary. Nonzero later is not, by itself, nonzero antecedent.

The burden is therefore symmetric. Zero cannot be promoted simply because it is standard or dynamically attractive, and nonzero cannot be promoted simply because a calculation can be initialized with it. Each claim has to survive at the same physical scope.

## 4. How physics can erase memory of an antecedent

There are strong physical reasons why a zero or effectively zero antecedent often appears natural in baryogenesis calculations. The most familiar begins with inflation. Exponential expansion strongly dilutes ordinary pre-existing particle densities, and the subsequent release of entropy during reheating can suppress a pre-existing baryon-to-entropy or baryon-to-photon ratio even further. In the standard inflationary picture this motivates the expectation that the observed baryon asymmetry must be generated after the principal entropy release.<sup>4</sup>

That is a powerful result within its physical domain. It is not yet the stronger statement that every admissible history has an exact zero baryon antecedent. Inflation is itself part of a cosmological architecture, and reheating is not a unique microscopic process. Even within inflationary models, the fate of an asymmetry depends on what carries it and how that carrier participates in inflation, reheating, and later interactions. The standard dilution argument therefore establishes severe suppression for broad classes of pre-existing number densities without automatically becoming an exact, carrier-independent zero theorem.

Thermal physics supplies another route toward forgetting initial conditions. Reactions that are rapid relative to the Hubble expansion impose equilibrium relations among chemical potentials, while baryon- and lepton-number-changing interactions can erase selected charge combinations. Electroweak sphalerons are especially important because they process baryon and lepton asymmetries in the high-temperature Standard Model. They do not, however, act as a generic eraser of every charge direction: their effect depends on the conserved combinations and on which other interactions are in equilibrium.<sup>5</sup>

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<sup>4</sup> A. H. Guth, “Inflationary Universe: A Possible Solution to the Horizon and Flatness Problems,” *Physical Review D* **23**, 347–356 (1981), doi:10.1103/PhysRevD.23.347; A. D. Dolgov and A. D. Linde, “Baryon Asymmetry in the Inflationary Universe,” *Physics Letters B* **116**, 329–334 (1982), doi:10.1016/0370-2693(82)90292-1; L. Kofman, A. Linde, and A. A. Starobinsky, “Towards the Theory of Reheating after Inflation,” *Physical Review D* **56**, 3258–3295 (1997), doi:10.1103/PhysRevD.56.3258. These sources provide the inflationary dilution and reheating context used in the discussion above.

<sup>5</sup> V. A. Kuzmin, V. A. Rubakov, and M. E. Shaposhnikov, *Physics Letters B* **155**, 36–42 (1985), doi:10.1016/0370-2693(85)91028-7; J. A. Harvey and M. S. Turner, “Cosmological Baryon and Lepton Number in the Presence of Electroweak Fermion-Number Violation,” *Physical Review D* **42**, 3344–3349 (1990), doi:10.1103/PhysRevD.42.3344. These sources describe electroweak sphaleron processing and the equilibrium relations among  $B$ ,  $L$ ,  $B - L$ , and chemical potentials.

Washout in leptogenesis makes the same point more sharply. In a strong-washout regime, an asymmetry present at sufficiently early times can be efficiently erased, and the final result can become largely insensitive to specified initial conditions. That is close to the kind of reset needed for a generation-from-zero argument. But even here the scope is physical rather than universal. Spectator fields can temporarily store asymmetry outside the interactions doing the washing out, partially equilibrated spectator processes can alter the final asymmetry, and sensitivity to initial conditions can reappear even in regimes conventionally described as strong washout.<sup>6</sup>

These are not weaknesses of inflation, equilibrium theory, or washout calculations. They are statements about what those calculations actually describe. Each can remove large regions of possible antecedent history. None thereby inherits authority over histories to which its physical assumptions do not apply.

Observations impose an independent set of restrictions. BBN and the CMB measure the baryon abundance through very different physics at widely separated epochs, and their agreement strongly constrains histories that would substantially alter the baryon-to-photon ratio between nucleosynthesis and recombination.<sup>7</sup> Likewise, the CMB strongly favors predominantly adiabatic primordial perturbations and restricts isocurvature contributions. Those constraints can exclude early histories that produce excessive spatial variation in the baryon asymmetry.<sup>8</sup>

But these observations perform a different job from an antecedent-zero theorem. They constrain what an admissible history may leave behind. Agreement between BBN and the CMB does not by itself reconstruct  $\Delta_B$  before BBN, and a tight bound on baryon isocurvature constrains spatial variation rather than fixing the homogeneous mean antecedent to zero.

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<sup>6</sup> B. Garbrecht and P. Schwaller, “Spectator Effects during Leptogenesis in the Strong Washout Regime,” *Journal of Cosmology and Astroparticle Physics* **2014**(10), 012 (2014), doi:10.1088/1475-7516/2014/10/012; S. Biondini et al., “Status of Rates and Rate Equations for Thermal Leptogenesis,” *International Journal of Modern Physics A* **33**, 1842004 (2018), doi:10.1142/S0217751X18420046. Garbrecht and Schwaller directly show that partially equilibrated spectators can hide asymmetry from washout and restore initial-condition sensitivity even in a strong-washout regime; Biondini et al. provides the broader thermal-rate context.

<sup>7</sup> R. J. Cooke, M. Pettini, and C. C. Steidel, *Astrophysical Journal* **855**, 102 (2018), doi:10.3847/1538-4357/aaab53; Planck Collaboration, N. Aghanim et al., *Astronomy & Astrophysics* **641**, A6 (2020), doi:10.1051/0004-6361/201833910. Together these sources provide the independent BBN- and CMB-based baryon-density determinations discussed in the text.

<sup>8</sup> Planck Collaboration, Y. Akrami et al., “Planck 2018 Results. X. Constraints on Inflation,” *Astronomy & Astrophysics* **641**, A10 (2020), doi:10.1051/0004-6361/201833887. This analysis places the relevant limits on primordial isocurvature contributions rather than treating exact adiabaticity as an assumption.



The zero-side case is therefore substantial. Together, these mechanisms and observations make many nonzero antecedent histories difficult or impossible. The remaining question is whether their reach is exhaustive. To establish generation from zero across  $H_{\text{adm}}$ , they would have to do more than eliminate broad subclasses: they would have to leave no physically admissible history with  $\Delta_B[C_B(h)] \neq 0$ .

That stronger conclusion does not follow from the existence of powerful erasure mechanisms alone. It requires exhaustion of the comparison class.

## 5. What a nonzero antecedent would have to survive

Failure to establish a universal zero antecedent does not establish a nonzero one. The opposite side of the comparison carries its own burden. A serious counterexample cannot consist merely of a cosmological calculation initialized with  $\Delta_B \neq 0$ . It must describe a physically coherent history in which the relevant nonzero net baryon asymmetry is genuinely antecedent at the comparison boundary, survives or is transmitted through the subsequent dynamics, and remains compatible with the observational constraints that define the admissible class.

There is nevertheless literature showing why initial-condition explanations cannot be dismissed simply by appealing to inflationary dilution. Krnjaic constructed an explicit scenario in which a large pre-inflationary asymmetry is stored in a light bosonic field and survives the enormous dilution associated with inflation through highly tuned initial conditions and couplings. The worked construction stores net  $B - L$ , with the asymmetry later communicated to the Standard Model after reheating. Its parameters are deliberately extreme: the field displacement is hyper-Planckian, the initial velocity must be chosen carefully, and interactions capable of transferring the asymmetry must avoid both excessive induced masses and destructive washout. The importance of the example is therefore not that it supplies a favored cosmology. It is that inflationary dilution alone is insufficient to constitute a theorem that all physically constructed antecedent asymmetries are erased.<sup>9</sup>

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<sup>9</sup> G. Krnjaic, “Can the Baryon Asymmetry Arise From Initial Conditions?,” *Physical Review D* **96**, 035041 (2017), doi:10.1103/PhysRevD.96.035041, arXiv:1606.05344. Krnjaic constructs a deliberately tuned inflationary history in which net  $B - L$  is stored in a light bosonic field with a hyper-Planckian initial displacement and transferred to the Standard Model after reheating. The paper’s narrow relevance here is important: it shows that inflationary dilution is not by itself a theorem against every initial-condition explanation, while the worked conserved quantity is  $B - L$  rather than the net- $B$  predicate used in this paper.

For the present question, however, that example also illustrates why the quantity under discussion cannot drift. A nonzero pre-inflationary  $B - L$  reservoir is not identical to a nonzero pre-inflationary net baryon asymmetry. Sphaleron and other transfer processes can later convert charge stored in one combination into baryon number, but such a history does not thereby establish  $\Delta_B \neq 0$  on the earlier boundary. A nonzero  $B - L$  or other charge reservoir therefore does not constitute a nonzero- $B$  antecedent merely because later sphaleron or transfer dynamics produce  $B \neq 0$  from it; when that conversion is part of the baryon-asymmetry-changing evolution under examination, the resulting  $B$  is downstream of  $C_B(h)$ , not antecedent to it. The Krnjaic construction is therefore a counterexample to an unrestricted claim of inflationary erasure, but its worked form is not yet the direct counterexample required by

$$\exists h \in H_{\text{adm}} \text{ such that } \Delta_B[C_B(h)] \neq 0.$$

The same distinction appears in a different way in wash-in leptogenesis. Domcke and collaborators showed that strong washout does not imply that every primordial asymmetry is physically irrelevant. Their construction allows nonzero chemical potentials in particle species that are not in equilibrium at the leptogenesis temperature; subsequent interactions reprocess this pre-existing charge information into a nonvanishing  $B - L$  asymmetry. The mechanism can operate precisely in a regime normally associated with strong washout.<sup>10</sup> This does not establish a nonzero antecedent  $B$ . It does show, however, that “washout erases the initial condition” is only meaningful after the relevant charge directions and equilibration network have been specified. Information stored outside the washout channel can remain physically consequential.

Initial-condition histories also face observational pressure that is considerably stronger than aesthetic objections about fine-tuning. Murai, Takahashi, Yamada, and Yin examined scenarios in which the amount of baryon asymmetry present in the later universe is already determined before inflation. Under the standard assumption that the observable curvature perturbation is subsequently generated by fluctuations of the

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<sup>10</sup> V. Domcke, K. Kamada, K. Mukaida, K. Schmitz, and M. Yamada, “Wash-In Leptogenesis,” *Physical Review Letters* **126**, 201802 (2021), doi:10.1103/PhysRevLett.126.201802, arXiv:2011.09347. The mechanism permits nonzero chemical potentials for species outside chemical equilibrium at the leptogenesis temperature and reprocesses those pre-existing charge asymmetries into nonzero  $B - L$ , including in a strong-washout regime. It therefore supplies a primary example in which strong washout does not imply erasure of every antecedent charge degree of freedom.

inflaton, variation in the duration of inflation makes the diluted baryon density fluctuate from patch to patch. In their treatment this produces a fully correlated baryon-isocurvature contribution of order  $S_B \simeq -3\mathcal{R}$ , far above the CMB limit for the mode considered. Broad pre-inflationary initial-asymmetry scenarios of that form are therefore observationally excluded, not merely disfavored.<sup>11</sup>

That result sharply narrows the counterexample space, but it does not close it. The assumptions responsible for the isocurvature signal are themselves physically identifiable. Murai and collaborators note that correlated cold-dark-matter isocurvature can compensate the baryon contribution, and they identify a second loophole when the baryon asymmetry is stored by the inflaton itself and is solely a function of the inflaton, in which case the relevant isocurvature terms are absent. They also distinguish baryogenesis occurring during inflation after observable CMB scales have crossed the horizon from the pre-inflationary case targeted by their exclusion.<sup>12</sup> These possibilities are not thereby established as viable explanations of the observed baryon asymmetry. They are enough to show that the isocurvature result is a powerful exclusion theorem for a defined history class rather than a theorem that every admissible antecedent must have  $\Delta_B = 0$ .

This is the point at which the symmetry of the evidentiary standard becomes consequential. It would be too permissive to take any formal initial condition with nonzero B as a counterexample. It would be equally restrictive to require a nonzero-antecedent history to explain the ultimate origin of its own boundary condition before it can enter the comparison. Standard generation-from-zero models also begin from physical conditions whose deeper origin may lie outside the model being calculated. The relevant requirement is instead that the history cover the causal interval being tested, make its baryon or charge bookkeeping explicit, and reach the established cosmological envelope without violating the constraints applicable to its own dynamics.

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<sup>11</sup> K. Murai, F. Takahashi, M. Yamada, and W. Yin, “Can baryon asymmetry be explained by a large initial value before inflation?,” *Physical Review D* **108**, 083518 (2023), doi:10.1103/PhysRevD.108.083518, arXiv:2307.03049; Planck Collaboration, Y. Akrami et al., “Planck 2018 Results. X. Constraints on Inflation,” *Astronomy & Astrophysics* **641**, A10 (2020), doi:10.1051/0004-6361/201833887. Murai et al. consider histories in which the later baryon abundance is already determined before inflation and show that, when the curvature perturbation is subsequently generated independently by the inflaton, the resulting baryon density carries a fully correlated isocurvature mode inconsistent with CMB constraints. The same paper explicitly records scope-limiting exceptions: correlated CDM isocurvature can compensate the baryon mode; if the baryon asymmetry is stored solely by the inflaton, the relevant isocurvature terms vanish; and baryogenesis during inflation after CMB scales leave the horizon is not excluded by the pre-inflation argument.

<sup>12</sup> Ibid.

Under that standard, the literature examined here does not yet supply the clean counterexample needed here. Physically articulated initial-condition and inherited-charge constructions demonstrate that inflation, reheating, and washout do not erase all conceivable antecedent information as a matter of theorem. At the same time, the strongest worked examples either carry quantities such as  $B - L$  rather than demonstrably nonzero  $B$  at the relevant boundary, incur significant observational constraints, or leave part of the required common antecedent scope outside the demonstrated model.

The result is therefore deliberately asymmetric with respect to certainty but symmetric with respect to method. Current physics does not establish

$$\forall h \in H_{\text{adm}}, \Delta_B[C_B(h)] = 0,$$

but neither does the presently examined literature establish

$$\exists h \in H_{\text{adm}}, \Delta_B[C_B(h)] \neq 0$$

at the same fully specified scope.

The antecedent-zero question remains open from both directions. That is not equivalent to ignorance about baryogenesis generally. Many mechanisms are well defined, many regions of parameter space are strongly constrained, and some antecedent histories are already excluded. What remains unresolved is the stronger scope claim: whether zero net baryon asymmetry is common to every physically admissible history relevant to the observed universe, or whether one physically complete nonzero- $B$  antecedent history survives the same evidentiary standard.

## 6. The scope test

The preceding comparison can be compressed into a simple conditional rule. Where the observed late nonzero asymmetry is retained and a zero antecedent has been established at physical scope  $S$ , the corresponding generation-from-zero burden becomes live at  $S$ .

If  $S$  denotes a declared physical scope, write

$$\text{ObservedNonzero} \wedge \text{ZeroEstablished}(S) \rightarrow \text{GenerationFromZeroBurden}(S).$$

The implication is intentionally one-directional. It does not say that a zero antecedent determines which baryogenesis mechanism operated, that a successful generation mechanism proves the antecedent was zero, or that observing a nonzero late asymmetry establishes anything about the earlier boundary. It says only that the observed nonzero datum and the established zero premise jointly make the corresponding generation-from-zero burden live at S.

That makes scope promotion an independent physical burden. If a model family M establishes

$$\text{ZeroEstablished}(M),$$

it does not follow that

$$\text{ZeroEstablished}(H_{\text{adm}}).$$

The missing step is not algebraic. It is physical. The evidence, theory, or applicable theorem has to describe the larger comparison class strongly enough to show that the zero condition survives the extension.

The strongest zero-side claim examined in this paper can therefore be written

$$\forall h \in H_{\text{adm}}, \Delta_B[C_B(h)] = 0.$$

Its falsifier is correspondingly direct:

$$\exists h \in H_{\text{adm}} \text{ such that } \Delta_B[C_B(h)] \neq 0.$$

One physically admissible history with a genuinely nonzero net-baryon antecedent at the relevant comparison boundary is sufficient to defeat the universal history-class claim. It would not show that this history was the actual history of the universe. It would show only that generation from zero cannot be promoted to the entire admissible class.

The converse is equally important. Failure to find such a history does not establish universal zero. The universal claim requires positive exhaustion of the admissible class: a theorem, observation, or demonstrably exhaustive combination of physical constraints strong enough to eliminate every admissible nonzero-B antecedent history.

“No known counterexample” is not logically equivalent to “all admissible histories have zero antecedent.”

This is the point at which the distinction between model enumeration and history-class coverage becomes decisive. No finite catalogue of familiar baryogenesis models can establish the universal statement merely because every model in the catalogue begins from an effectively symmetric state. That result would apply to the catalogue. To extend it to  $H_{\text{adm}}$ , one must additionally show that the catalogue exhausts every physically admissible history relevant to the comparison.

The accompanying formal contract encodes this conditional structure without selecting the physical history. The empirical predicates—what counts as an admissible history and whether its antecedent net baryon asymmetry is zero—remain supplied by physical description. The machine-checkable layer tests the inference once those predicates are specified. It therefore cannot turn an open cosmological premise into an empirical result.

## **7. What present experiments can—and cannot—decide**

The absence of a universal antecedent-zero result does not imply that current observations and experiments are uninformative. On the contrary, several experimental programs cut deeply into the admissible history space. Their limitation is one of reach: they discriminate properties of histories and mechanisms without presently determining the antecedent-zero predicate across the full comparison class.

CMB measurements provide the clearest example. Limits on baryon and related isocurvature modes can exclude histories in which a pre-existing asymmetry produces excessive spatial variation. As discussed above, this already places strong pressure on broad classes of pre-inflationary baryon-asymmetry scenarios. But an isocurvature measurement constrains variation around the mean. Even an arbitrarily tight upper bound on one isocurvature mode does not, by itself, establish that the homogeneous antecedent mean satisfies  $\Delta_B = 0$ .

Searches for cosmological antimatter constrain a different dimension of the problem. A universe partitioned into large matter and antimatter domains would produce annihilation at their boundaries and associated diffuse gamma-ray and CMB signatures. Cohen, De Rújula, and Glashow showed that a globally

matter-antimatter-symmetric domain structure of this type is incompatible with the observed diffuse background unless the matter domain containing us occupies essentially the visible universe.<sup>13</sup> This is a strong constraint on spatially sign-varying histories. It does not distinguish a homogeneous positive baryon excess generated dynamically from a homogeneous positive excess inherited as an antecedent condition.

Laboratory searches for baryon-number violation probe another necessary ingredient of broad classes of baryogenesis models. Neutron-antineutron oscillations would provide direct evidence for  $\Delta B = 2$  baryon-number violation; the NNBAR program at the European Spallation Source, developed from the earlier HIBEAM/NNBAR concept, targets a substantial improvement in sensitivity to such conversions.<sup>14</sup>

A discovery in one of these channels could materially change the baryogenesis landscape. It could establish a previously unobserved baryon-number-violating interaction, eliminate or strengthen mechanism families, and alter which early-universe histories remain physically credible. A null result can likewise close parameter regions. But neither outcome directly supplies the value of  $\Delta_B$  on a cosmological antecedent boundary. A laboratory interaction demonstrates what particle physics can do under the conditions probed; translating that result into a statement about the realized early-universe history requires additional cosmological structure.

Other measurements can likewise shrink  $H_{\text{adm}}$  without directly determining the antecedent. A sufficiently rich combination could in principle become decisive. The requirement is not that one spectacular experiment measure the antecedent directly. The requirement is that the surviving physical possibilities become exhaustive in the relevant sense: either every remaining admissible history shares  $\Delta_B[C_B(h)] = 0$ , or at least one fully admitted history demonstrably does not.

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<sup>13</sup> A. G. Cohen, A. De Rújula, and S. L. Glashow, “A Matter-Antimatter Universe?,” *The Astrophysical Journal* **495**, 539–549 (1998), doi:10.1086/305328, arXiv:astro-ph/9707087. The authors analyze a cosmology divided into matter and antimatter domains and calculate the annihilation signatures expected at domain boundaries, including contributions to the diffuse gamma-ray background and distortions of the CMB. They conclude that such a matter-antimatter-symmetric domain universe is empirically excluded unless the matter region containing us is effectively the entire visible universe.

<sup>14</sup> A. Addazi et al., “New high-sensitivity searches for neutrons converting into antineutrons and/or sterile neutrons at the HIBEAM/NNBAR experiment at the European Spallation Source,” *Journal of Physics G: Nuclear and Particle Physics* **48**, 070501 (2021), doi:10.1088/1361-6471/abf429, arXiv:2006.04907. The proposed HIBEAM/NNBAR program targets free neutron-antineutron conversion and related neutron-sector processes, including  $\Delta B = 2$  baryon-number violation, with sensitivity substantially beyond previous free-neutron searches. Its relevance here is mechanism-level: observation of baryon-number violation would establish an important physical ingredient for classes of baryogenesis models without determining the cosmological antecedent value of  $\Delta_B$ .

No finite set of observations or experiments identified in the literature examined here has been shown to exhaust  $H_{\text{adm}}$  in that sense.

This makes the current experimental disposition more specific than saying that baryogenesis remains an open problem. Mechanism searches remain active and scientifically discriminating. Cosmological observations already exclude important antecedent histories. What is missing for the question posed here is a distinctive adjudicator of the stronger history-class statement.

At present, therefore, the zero antecedent remains unestablished across  $H_{\text{adm}}$ , while the nonzero counterexample condition also remains physically unfulfilled at the same scope. The formal falsifier exists. The decisive physical witness—or an exhaustive theorem excluding every such witness—does not.

That is the current adjudication.

## 8. Interpretation

The result does not remove the baryogenesis problem. It changes the scope at which one version of that problem can be stated.

Within a history or model family for which  $\Delta_B = 0$  has been physically established at the relevant antecedent boundary, the generation-from-zero question remains exactly as strong as usual. Such a history must describe how a later nonzero baryon asymmetry can arise from that antecedent, and the applicable dynamical requirements remain in force. Nothing in the scope argument weakens a baryogenesis calculation inside the domain for which its premises hold.

What does not follow is the promotion of that burden to every physically admissible history merely because the observed universe has  $\Delta_B \neq 0$ . The observation fixes a late condition. Inflation, reheating, equilibrium relations, sphalerons, washout, and cosmological data constrain possible routes to that condition. None of those facts substitutes for establishing the antecedent at the broader history-class scope.

The title's "missing premise" is therefore literal rather than rhetorical. The premise is not that baryogenesis mechanisms are implausible, nor that a primordial asymmetry



should be preferred. It is the zero antecedent itself when the explanatory claim is enlarged from a specified physical history to the full admissible comparison class.

This distinction also changes what a successful future result would look like. One route would be a physical theorem whose assumptions are themselves satisfied throughout  $H_{\text{adm}}$  and which drives net baryon asymmetry to zero before the relevant generation dynamics begin. Another would be an observationally exhaustive partition of the admissible histories in which every nonzero-antecedent branch is excluded. Either would close the zero side.

The opposite result is cheaper in logical form but not in physical content. A single history satisfying

$$\Delta_B[C_B(h)] \neq 0$$

would defeat all-history zero only if that history also survives the same admissibility standard used throughout this paper. It must cover the relevant antecedent interval rather than beginning after an unmodeled generation event; it must track  $B$  rather than substituting  $B - L$  or another charge without the required conversion; and it must remain compatible with the relevant cosmological and particle-physics constraints. No such history has been established here.

This leaves three questions that should not be conflated. Whether a particular baryogenesis mechanism can generate the observed asymmetry is a model-level physical question. Whether every admissible history begins from the relevant zero state is a history-class question. Which history the observable universe actually realized is a third question again. A result at one level does not answer the others without an additional physical bridge.

The formal contract accompanying this paper is intended only to preserve those distinctions. It does not assign the antecedent, choose a mechanism, or determine cosmic history. Its role is narrower: once a physical scope and antecedent are supplied, the conditional relation between that premise and the corresponding generation-from-zero burden is checked explicitly. The empirical content remains with physics.

## 9. Conclusion

The observed matter-antimatter asymmetry is a late cosmological fact. Its antecedent is not observed in the same way. Inflationary dilution, reheating, thermal processing, washout, and cosmological constraints strongly restrict possible earlier histories, but they do not establish an exact zero net-baryon antecedent across every physically admissible history. Conversely, the strongest nonzero-antecedent examples examined here do not yet supply a fully admitted history with demonstrably nonzero net baryon asymmetry at the common antecedent boundary.

The present result is therefore not a choice between a symmetric and an asymmetric beginning. It is a scope result:

$$\forall h \in H_{\text{adm}}, \Delta_B[C_B(h)] = 0$$

has not been established, while

$$\exists h \in H_{\text{adm}} \text{ such that } \Delta_B[C_B(h)] \neq 0$$

has not been physically established either.

Accordingly, generation from zero should be treated as a scoped physical burden. Where a zero antecedent is established, the burden is live. Where the zero premise is restricted to a model family, the burden is restricted with it. Promotion to the full admissible-history class requires an additional physical result establishing that the zero condition survives the promotion. The claim remains falsifiable in both directions: an exhaustive zero result across  $H_{\text{adm}}$  would close the burden at that scope, while one fully admissible nonzero-B antecedent history would defeat the universalization without establishing that the history was actual.

That is the missing premise. At present, the appropriate conclusion is narrower than a new baryogenesis mechanism and more specific than the statement that baryogenesis remains unsolved:

**the observed baryon asymmetry does not presently carry with it an established zero antecedent at the full physically admissible history-class scope.**

## **AI Use Disclosure**

Artificial intelligence tools were used during the development of this work for research assistance, adversarial review, conceptual organization, drafting support, editing, and formalization support. The author directed the research program, selected and evaluated the scientific claims and sources, determined the argument and its scope, and takes responsibility for the final content.

# The Missing Premise in Baryogenesis: The Scope of the Zero Antecedent

Signed,  
  
Rick